# Time Series Analysis Assignment Report

## 1. Data

The dataset consists of quarterly data collected from FRED on price level, unemployment rate and the Federal Funds Rate, from the years 1960 – 2025. The inflation rate was then calculated using the formula:

$$\pi_t = 400 \ln(P_t/P_{t-1}))$$

## 2. Optimal Lags

To understand the lags which would be optimal for the purpose of this exercise, we conducted the following tests and plotted the following graphs.

### a. Partial Auto-correlation Function (PACF) plots

A PACF plot helps us understand the correlation of a variable to its lag while isolating the direct effect of each lag by removing the effect of other lags. This helps us deduce the lags that should be included in our analysis models.
In our graphs, the lines crossing the significance threshold (blue band) are lags T-1, T-2 and T-3 depicting that these 3 months have the largest explanatory power for our variable.

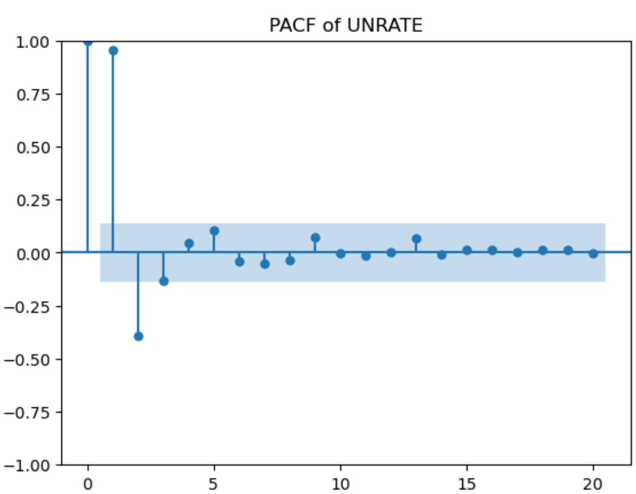


*Fig1: PACF Plot for unemployment rate*

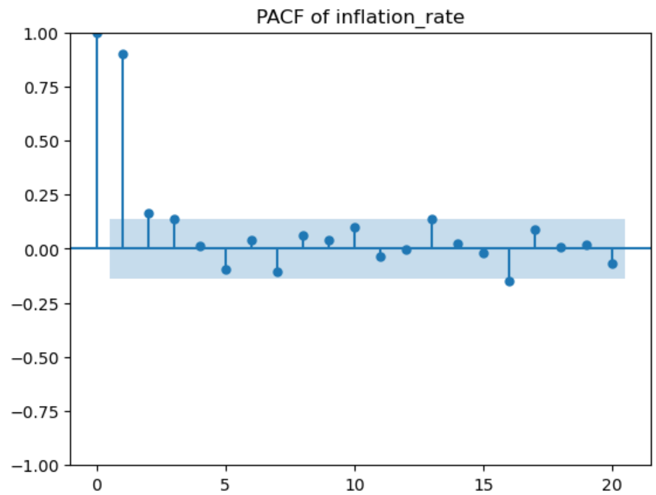


*Fig 2: PACF Plot for Inflation rate*

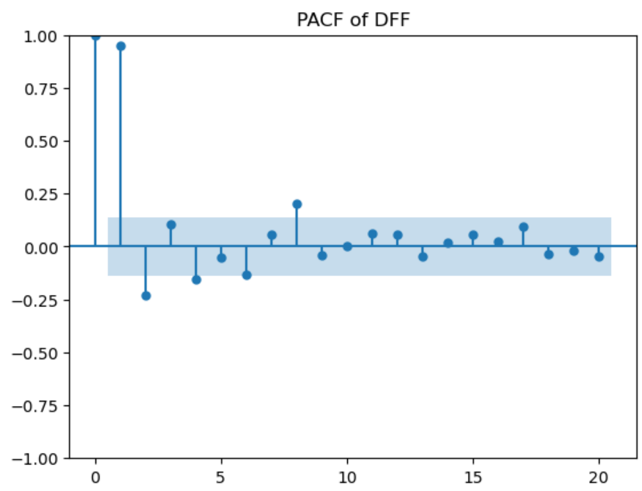


*fig 3: PACF of Fed Funds Rate*

**b. <u>ar_select_order method</u>**

The ar_select_order function on Python returned the following lags as the optimal lags:

```
inflation_rate: selected lags = [1, 2, 3]
UNRATE: selected lags = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
DFF: selected lags = [1, 2, 3, 4, 5, 6, 7, 8]
```

*Fig 4: optimal lags as per ar_select_order function*

The output does not match the PACF plots completely. This method takes into consideration the T-8$^{th}$ lag for the Fed Funds Rate thereby overstating the significant lags for the variable. The T-9$^{th}$ lag, as seen in fig 3, does indeed cross the significance threshold, but is too isolated and far back to be taken into consideration. Similarly, it overestimates the optimal lags for unemployment rate. This method, however, aptly measures the optimal lag for inflation rate.

Therefore for analyzing the time series I will be taking lags upto T-3 into account which match the results of the PACF plot.

3. **<u>Forecasting on test data</u>**

Using the optimal lags and train data, the following forecasts of the test data.

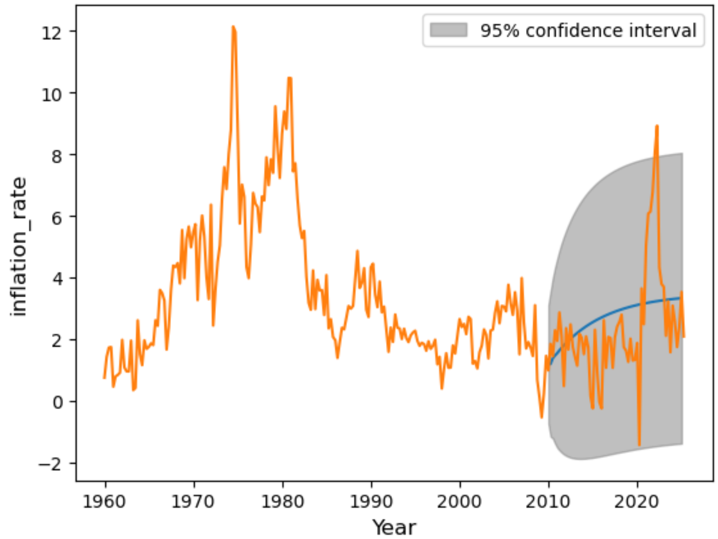


*Fig 5: AR forecast for inflation rate*

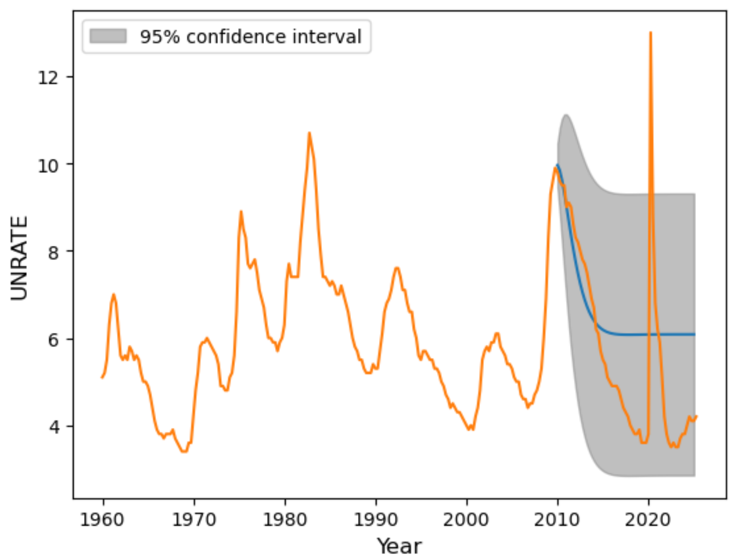


*Fig 6: AR forecast for unemployment rate*

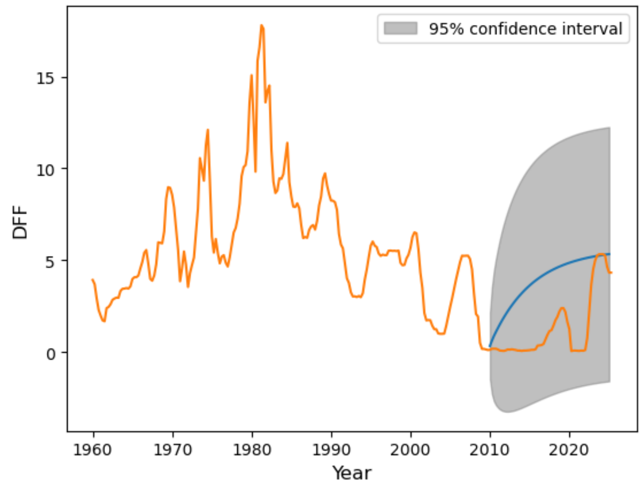


*Fig 7: AR forecast for Fed Funds rate*

The forecast matches the trend of our test data across all three variables but does not capture any volatility across 2010-2025. I think this forecast is similar to what I expected given that I knew I wasn't adding any volatility measures.

## 4. Forecasting Forward

Using parameters from the previous model to forecast rates up to 2027, yield the following results:

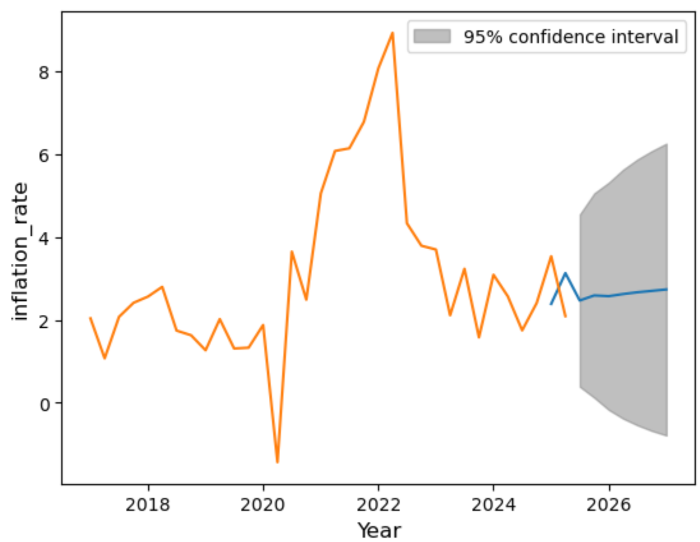


*Fig 8: 2025-2027 forecasts of inflation rate*

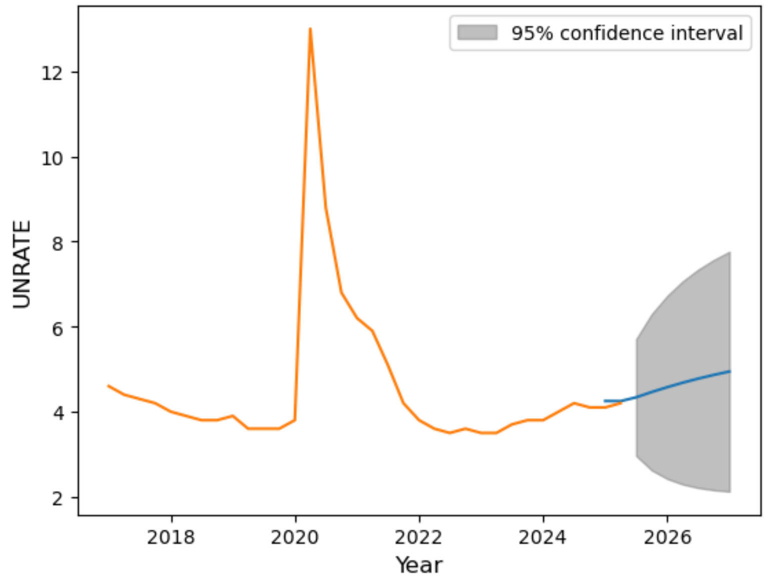


*Fig 9: 2025-2027 forecasts of unemployment rate*

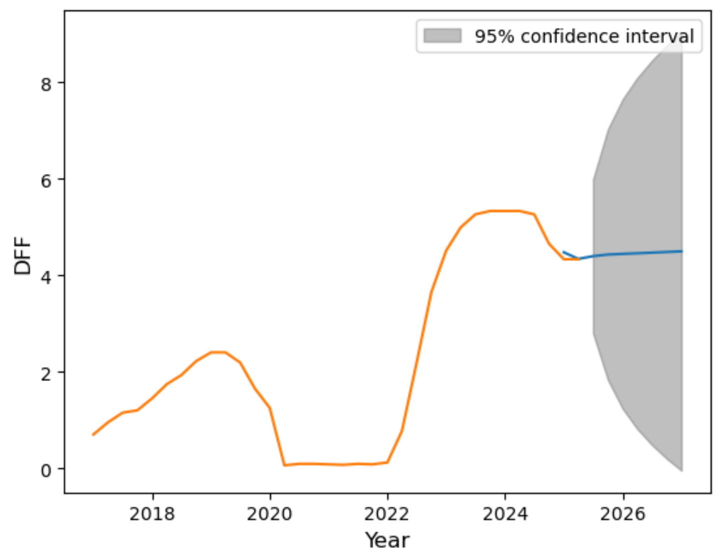


*Fig 10: 2025-2027 forecasts of Fed Funds rate*

Again, the forecasts roughly follow the trend of the last two years for all three rates. These graphs are unable to capture and predict any volatility, and are not the strongest forecasts.

5. **<u>Estimating the VAR model (Sample: 1960-2000)</u>**
   a. **<u>Impulse Response Functions:</u>**

   Replicating the paper, following are the impulse response functions for the unemployment rate, inflation rate and Fed Funds rate from the years 1960 to 2000.

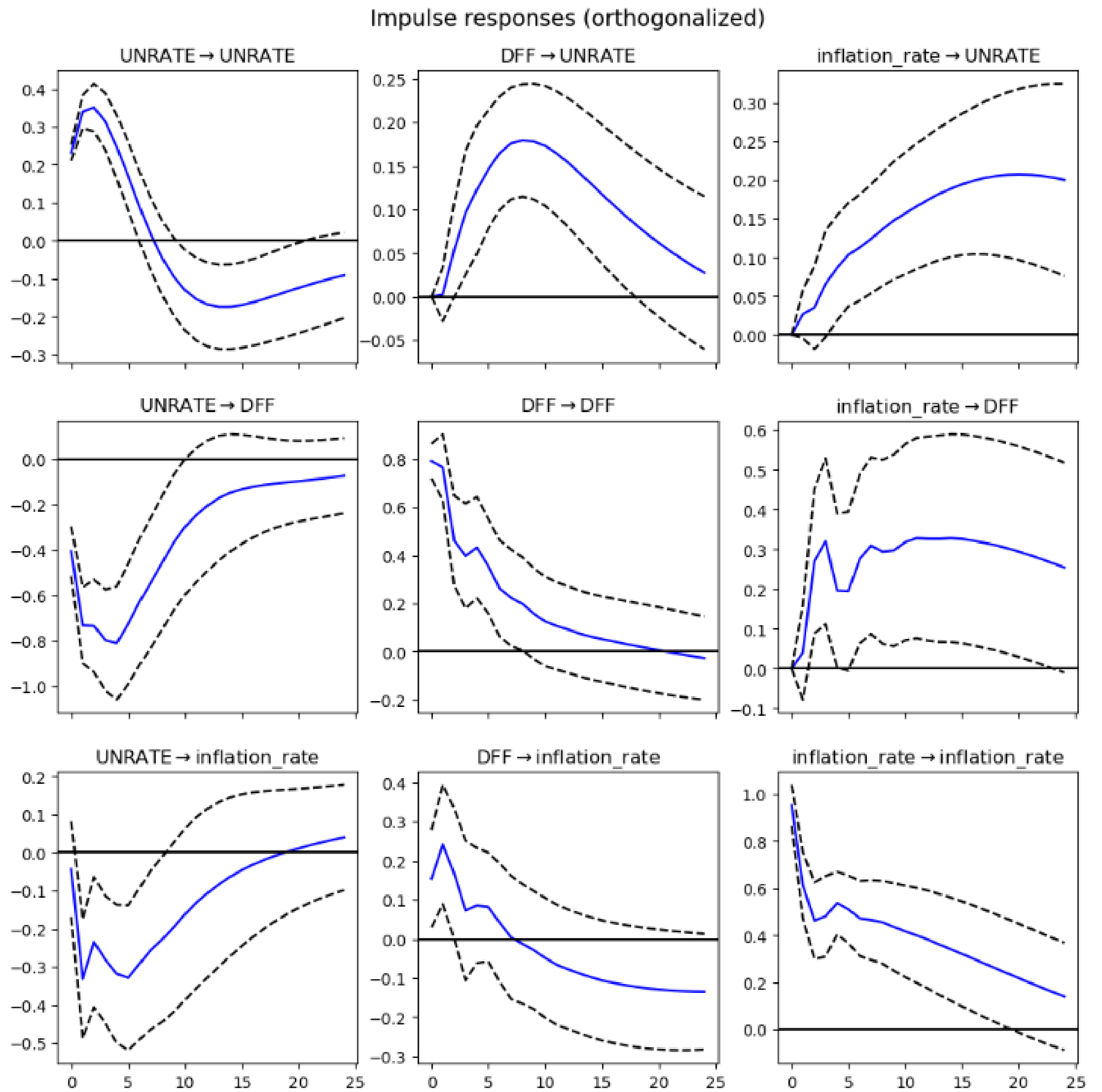


*Fig 11: Impulse Response Functions*

Key interpretations are as follows:

<u>Monetary Policy (DFF Shock):</u> An unexpected interest rate increase leads to higher unemployment and lower inflation. This is the standard mechanism for cooling the economy.

<u>Central Bank Reaction:</u> The central bank cuts rates when unemployment rises and raises rates when inflation rises. This shows a systematic response to its dual mandate.

<u>Inflation-Unemployment Trade-off (Phillips Curve):</u> A rise in unemployment causes inflation to fall. Conversely, a rise in inflation temporarily lowers unemployment. This shows a short-run trade-off.

Shocks to a variable (e.g., a spike in unemployment or inflation) have persistent effects, meaning they do not disappear immediately.

## b. <u>Forecast Error Variance Decomposition</u>

FEVD shows what percentage of the forecast uncertainty for each variable is caused by shocks to itself versus the other variables over time.

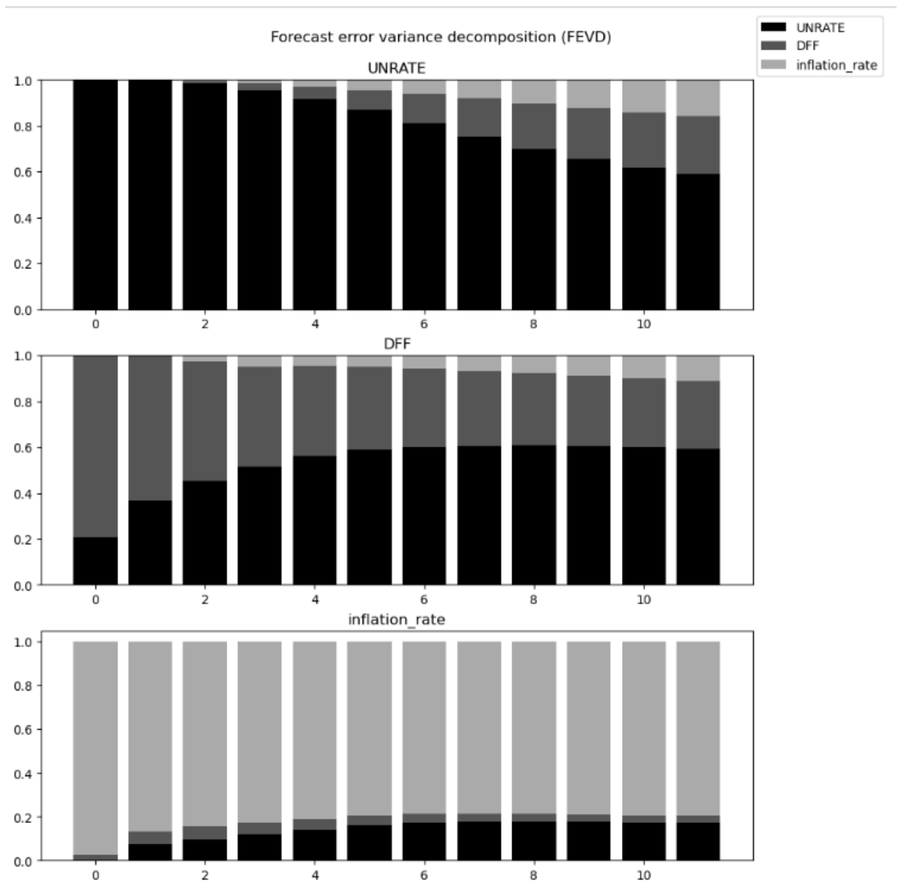


*Fig12: FEVD Plot*

UNRATE (Unemployment Rate):

- Most of its own future uncertainty is explained by shocks to itself, especially in the short term.
- Shocks from the interest rate (DFF) and inflation play a smaller but growing role over the 10-period horizon.

DFF (Interest Rate):

- The vast majority of its future variation is explained by its own shocks.

- Shocks from unemployment and inflation have very little influence. This suggests the interest rate is largely driven by its own policy dynamics or external factors not captured by this model.

Inflation_rate:

- Like DFF, its own future uncertainty is predominantly caused by shocks to itself.
- Shocks from unemployment and the interest rate explain only a minor part of its forecast variance.

c. **Granger Causality Tests**

Results for the tests are as follows

| Regressor | Inflation Rate | Unemploment Rate | Fed Funds Rate |
|---|---|---|---|
| Inflation Rate | 0.00 | 0.422 | 0.000 |
| Unemployment Rate | 0.020 | 0.00 | 0.000 |
| Fed Funds Rate | 0.509 | 0.002 | 0.00 |

*Fig 13: Granger Causality Test Results*

Using the threshold of p-value > 0.05 the following pairs are Granger causal:

- Unemployment Rate → Inflation Rate
- Unemployment Rate → Fed Funds Rate
- Inflation Rate → Fed Funds Rate
- Fed Funds Rate → Unemployment Rate

While p-vales in this test are different, the overall causal pair are the same as Stock and Watson (2001).

## 6. Estimating the VAR model (Sample: 1960-2000)

Replicating the steps in part 5 to spot differences across time

a. **Impulse Response Functions:**

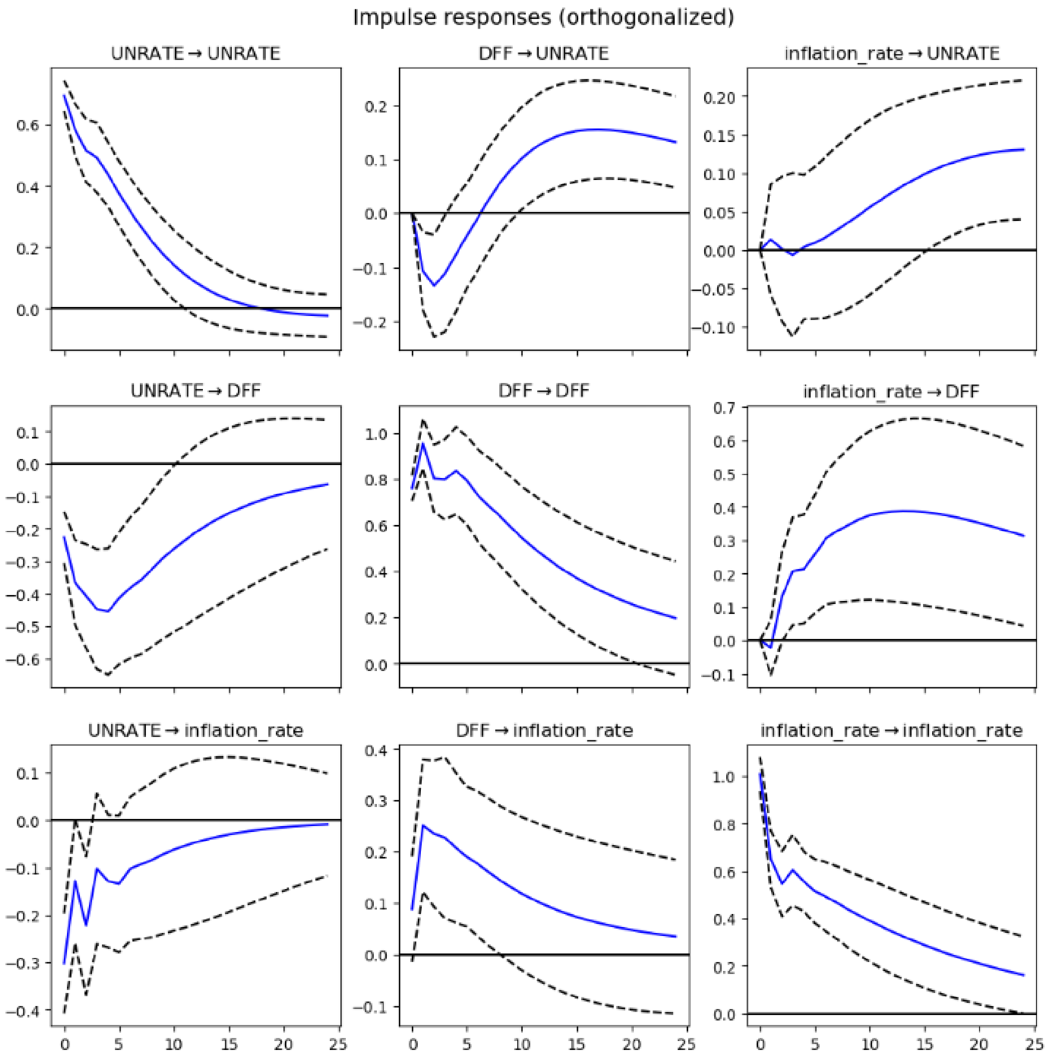


*Fig 14: Impulse Response Functions*

<u>Monetary Policy Shock:</u> The effect on unemployment (DFF → UNRATE) is positive but very small and short-lived. The effect on inflation (DFF → inflation_rate) is immediately and persistently negative. This is a clear price puzzle-like effect, showing that a rate hike successfully lowers inflation.

<u>Inflation Shock:</u> Leads to a slight decrease in unemployment (inflation_rate → UNRATE), consistent with a short-run Phillips Curve. Provokes a strong, persistent increase in interest rates (inflation_rate → DFF), showing a vigorous central bank response.

<u>Unemployment Shock:</u> Leads to a small, transient drop in inflation (UNRATE → inflation_rate). The response of interest rates (UNRATE → DFF) is ambiguous and very close to zero, showing no clear policy reaction.

**Comparison with Previous Sample (1960-2000):**

- **Monetary Policy is More Effective:** The most critical difference is in the DFF → inflation_rate response. In the earlier sample, this effect was statistically insignificant. Now, it is clearly negative and significant, indicating that monetary policy has a much more reliable and identifiable effect on controlling inflation in the modern era.
- **Weaker Phillips Curve:** The responses between unemployment and inflation (UNRATE ↔ inflation_rate) are weaker and less persistent than one might expect from the 1960-2000 period, supporting the "flattening of the Phillips Curve" narrative.
- **Less Reactive Central Bank to Unemployment:** The UNRATE → DFF response is now virtually zero, whereas before it was negative (the Fed cut rates). This suggests that in the modern era, the central bank's reaction function may be more focused on inflation than on unemployment.

## b. <u>Forecast Error Variance Decomposition</u>

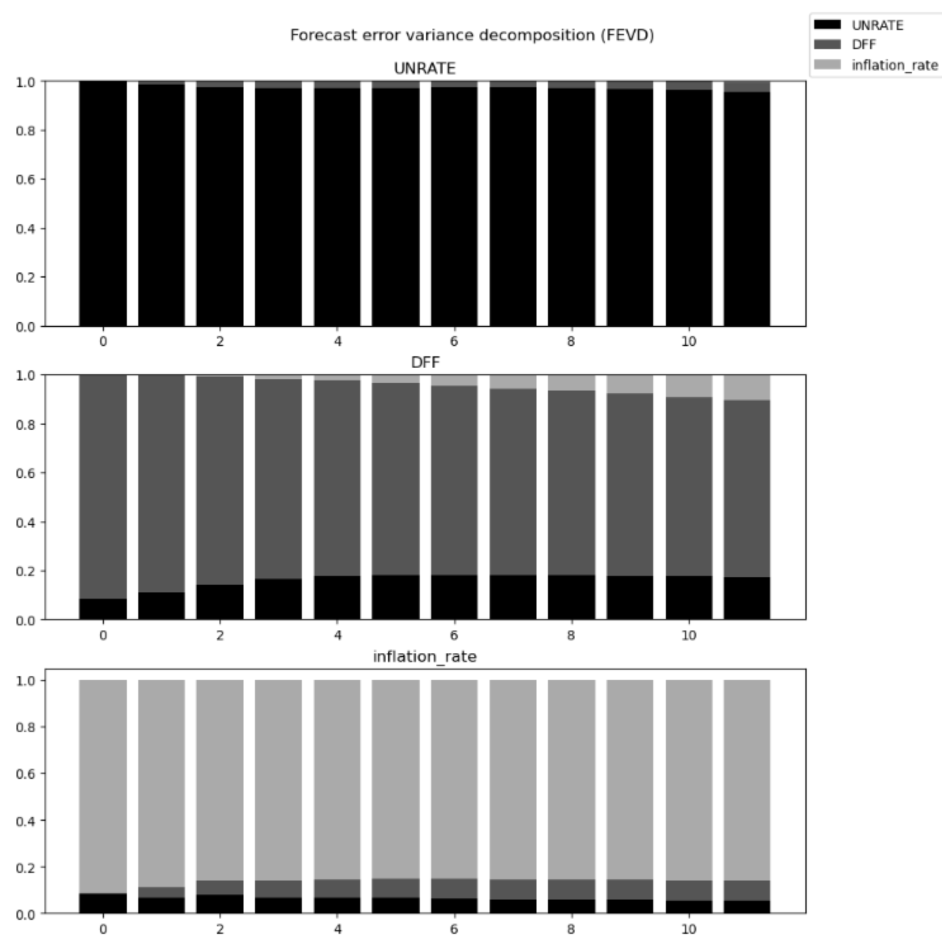


*Fig15: FEVD Plot*

**Interpretation (1960-2025):** The graphs show that for all three variables, the vast majority of their forecast error variance (the dark, solid lines are very high) is explained by their own shocks. The contributions from other variables are minimal.

**Comparison with Previous Sample (1960-2000):** In the earlier sample, there was a visible, though minor, growing influence of other variables on UNRATE over time. In the new FEVD graphs, this external influence appears even smaller. This reinforces the causality results: the variables have become more independent of each other. The system is more decoupled, meaning shocks in one variable explain very little of the future uncertainty in another.

## c. Granger Causality Tests

| Regressor | Inflation Rate | Unemploment Rate | Fed Funds Rate |
|---|---|---|---|
| Inflation Rate | 0.00 | 0.896 | 0.005 |
| Unemployment Rate | 0.416 | 0.00 | 0.412 |
| Fed Funds Rate | 0.044 | 0.013 | 0.00 |

*Fig 16: Granger Causality Test Results*

**Interpretation (1960-2025):**

Using the threshold of p-value > 0.05 the following pairs are Granger causal:

- Fed Funds Rate → Inflation Rate
- Fed Funds Rate → Unemployment Rate
- Inflation Rate → Fed Funds Rate

**Comparison with Previous Sample (1960-2000):**
This is a complete reversal. In the 20th-century data, the story was that the labor market (unemployment) drove the system, causing changes in both inflation and policy. In the full sample including 21st-century data, that relationship has broken down. Now, monetary policy (DFF) appears as a key predictive force for inflation, while unemployment's predictive power has faded. This aligns with the idea that the Phillips Curve relationship (the inflation-unemployment trade-off) has weakened or become less stable, while central bank credibility and policy actions have become more influential for inflation dynamics.